NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Marks: 100	**ASSIGNMENT-3**	**Dead Line: 15-03-26**

Subject: Probability and Statistics **Sections: ALL**
Instructor: Mr. Hazber Samson, Miss Javeria, Miss Warda Date: 12-03-2026

Q1. The table below shows the probability distribution for a random variable X.

x	0	1	2	3
$P(X=x)$	c	c^2	c^2+c	$3c^2+2c$

(a) Find the value of *c.*
(b) Find distribution function $F(X)$ i.e. cdf.
(c) Find $E(X)$
(d) Find $V(X)$
(e) Find $E(2X+3)$ *and* $V(5X+4)$

Q2. A number is picked at random from the set of integers $\{1,2,3,\ldots,100\}$. Suppose X represents the remainder after dividing the number by 7.
(a) Find the probability distribution of X.
(b) $P(1\le X\le 5)$ (c) $P(X\le 3)$ (d) $P(X\ge 4)$ (e) $P(|X|\le 1)$

Q3. A random variable X has probability density function $f(x)$ where

$$f(x)=\begin{cases} kx & 0\le x<1 \\ k(2-x) & 1\le x\le 2 \\ 0 & otherwise \end{cases}$$

Then find
(a) $P(0.5\le X\le 1.5)$
(b) $E(X)$
(c) $V(X)$

Q4. A continuous random variable X has p.d.f $f(x)$ where

$$f(x)=\begin{cases} A(2-x)(2+x) & 0\le x\le 2 \\ 0 & otherwise \end{cases}$$

Find the following
(a) value of constant *A*
(b) $P(X\le 1)$
(c) $P(X\ge 2)$
(d) $P(1<X<2)$
(e) *Median and Mode*

Q5. The random variable *X* can take only the values -2, -1, 0, 1, 2. The probability distribution of *X* is given in the following table.

x	-2	-1	0	1	2
$\mathrm{P}(X=x)$	p	p	0.1	q	q

(a) Given that $P(X \geq 0) = 3P(X < 0)$, find the values of *p* and *q*.

(b) Find distribution function $F(X)$ for X also find $P(X \leq 1)$ using **cdf**.

(c) Find $E(X)$

(d) Find $V(X)$

(e) Find $E(5X+3)$ *and* $V(2X+1)$

Q6. Let *X* denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let *Y* denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as

$f(x,y)$		x = 1	x = 2	x = 3
y	1	0.05	0.05	0.10
	3	0.05	0.10	0.35
	5	0.00	0.20	0.10

(a) Evaluate the marginal distributions of *X* and Y.

(b) Find *P*(*Y* = 3 | *X* = 2), $P(X > Y)$, $P(X+Y \leq 3)$

(c) Check whether X and Y are independent or not.

(d) *Find* $E(X+Y)$, $E(X-Y)$, $E(XY)$, $E(X)$, $E(Y)$

(e) *Find* $COV(X,Y)$ *and* $Corr(X,Y)$

Q7. From a sack of fruit containing 3 oranges, 2 apples, and 3 bananas, a random sample of 4 pieces of fruit is selected. If *X* is the number of oranges and *Y* is the number of apples in the sample, find

(a) the joint probability distribution of *X* and *Y* ;

(b) $P(x+y \leq 2)$

(c) Check whether X and Y are independent or not.

(d) *Find* $E(XY)$, $E(X)$, $E(Y)$

(e) *Find* $COV(X,Y)$ *and* $Corr(X,Y)$

Q8. Consider the following joint probability density function of the random variables *X* and *Y* :

$$f(x,y) = \begin{cases} \dfrac{3x-y}{9} & 1 < x < 3,\ 1 < y < 2 \\ 0 & \textit{elsewhere} \end{cases}$$

(a) Find the marginal density functions of *X* and *Y* .

(b) Find *P*(1.5<*X*<2.5, 0.5<Y<1.5). $P(X > 2)$, $P(Y \leq 1.5)$

(c) Find Conditional probability functions and check whether *X* and *Y* independent or not?

(d) $P(X > Y)$, $P(X+Y < 1)$

(e) *Find* $COV(X,Y)$ *and* $Corr(X,Y)$

Q9. The random variables X and Y are jointly distributed as

$$f(x,y)=\begin{cases}24x^2y(1-x) & 0\le x,y\le 1\\ 0 & elsewhere\end{cases}$$

Find
(a) marginal distributions of X
(b) conditional distribution of Y.
(c) Are X and Y independent?
(d) *Find* $E(XY)$, $E(X)$, $E(Y)$
(e) *Find* $COV(X,Y)$ *and* $Corr(X,Y)$

Q10. Suppose that in a certain drug the concentration of a particular chemical is a random variable with the following continuous distribution

$$g(x)=(3/8)x^2 \quad for \quad 0\le x\le 2 \ \ \& \ 0 elsewhere$$

Suppose that the concentrations X and Y of the chemical in two separate batches of the drug are independent random variables each with the same p.d.f . Determine
(a) the joint p.d.f of X and Y
(b) $P(X>Y)$
(c) $P(X+Y\le 1)$